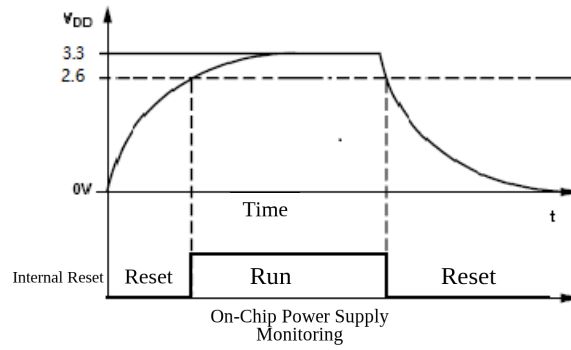
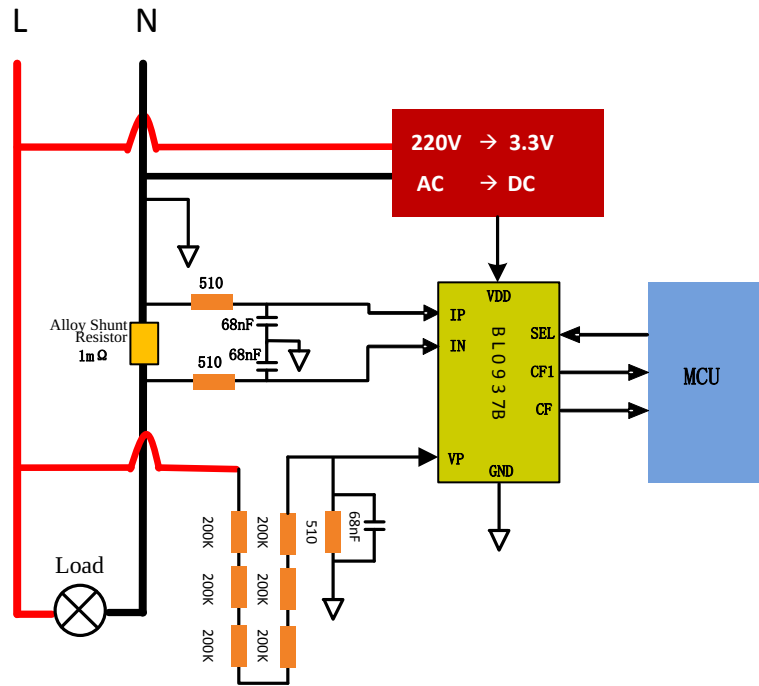




8 Applications

8.1 Typical BL0937B Application

A typical application block diagram for the BL0937B is shown below. The device operates from a 3.3V supply. The current signal is sensed across an alloy shunt resistor and applied to the IP and IN pins of the BL0937B, while the voltage signal is attenuated by a resistive divider and applied to the VP pin. The CF, CF1, and SEL pins connect directly to MCU pins. Active power, RMS current, and RMS voltage are calculated by measuring the CF and CF1 pulse periods.



8.2 CF and CF1 Output Frequencies

The BL0937B multiplies the input voltages of the voltage and current channels and processes the resulting signal to convert the active-power information into a frequency. At the same time, it calculates the RMS voltage and RMS current and converts them into corresponding frequencies. The resulting active-high pulse trains are output at CF for active power and at CF1 for RMS voltage or current.

(1) Active-power output pulse frequency:

$$F_{CF} = 1721506 \times \frac{V(V) \times V(I)}{V_{ref}^2}$$

(2) RMS-voltage output pulse frequency: